

Python - Nested Loops

In Python, when you write one or more **loops** within a loop statement that is known as a **nested loop**. The main loop is considered as outer loop and loop(s) inside the outer loop are known as inner loops.

The Python programming language allows the use of one loop inside another loop. A **loop** is a code block that executes specific instructions repeatedly. There are two types of loops, namely for and while, using which we can create nested loops.

You can put any type of loop inside of any other type of loop. For example, a for loop can be inside a while loop or vice versa.

Python Nested for Loop

The **for loop** with one or more inner for loops is called nested for loop. A for loop is used to loop over the items of any sequence, such as a list, tuple or a string and performs the same action on each item of the sequence.

Python Nested for Loop Syntax

The syntax for a Python **nested for loop** statement in Python programming language is as follows –

```
for iterating_var in sequence:
    for iterating_var in sequence:
        statements(s)
    statements(s)
```

Python Nested for Loop Example

The following program uses a nested for loop to iterate over months and days lists.

```
months = ["jan", "feb", "mar"]
days = ["sun", "mon", "tue"]

for x in months:
    for y in days:
        print(x, y)

print("Good bye!")
```

When the above code is executed, it produces following result –

```
jan sun
jan mon
jan tue
feb sun
feb mon
feb tue
mar sun
mar mon
mar tue
Good bye!
```

Python Nested while Loop

The **while loop** having one or more inner while loops are nested while loop. A while loop is used to repeat a block of code for an unknown number of times until the specified boolean expression becomes TRUE.

Python Nested while Loop Syntax

The syntax for a **nested while loop** statement in Python programming language is as follows –

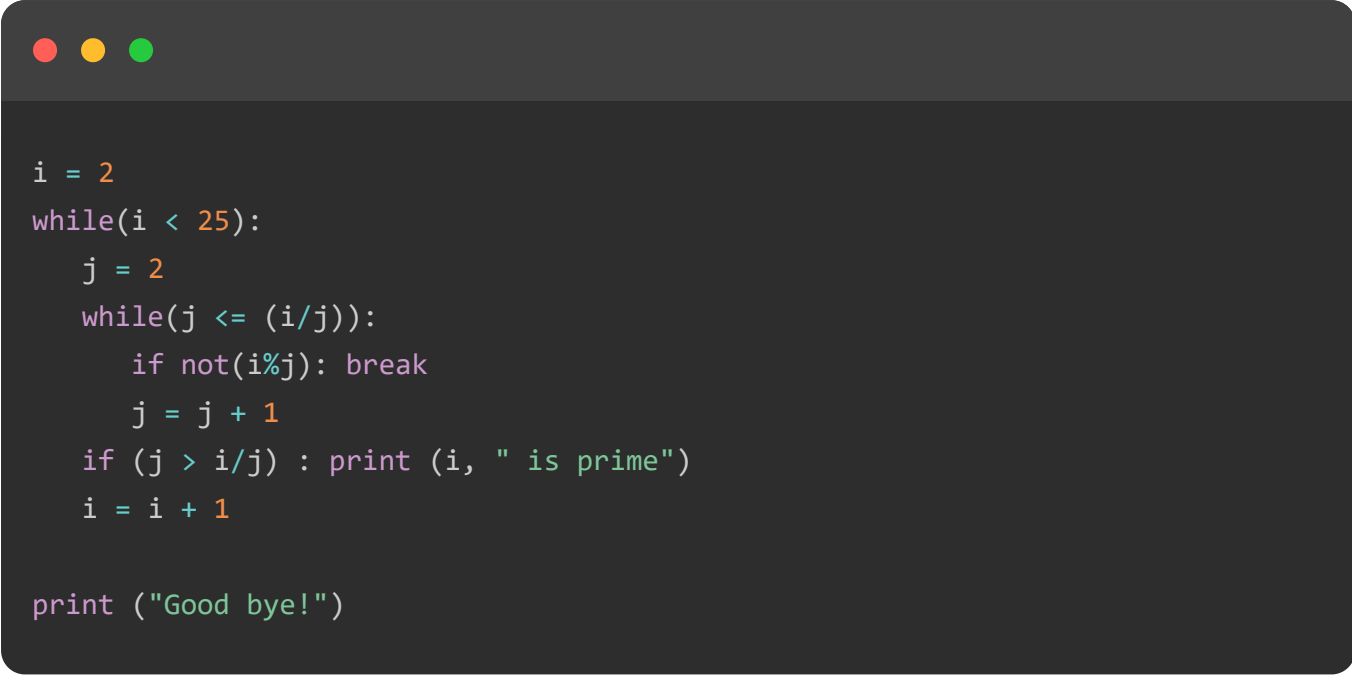
```
while expression:
    while expression:
```

```
statement(s)
statement(s)
```

Python Nested while Loop Example

The following program uses a nested while loop to find the prime numbers from 2 to 100

–



```
i = 2
while(i < 25):
    j = 2
    while(j <= (i/j)):
        if not(i%j): break
        j = j + 1
    if (j > i/j) : print (i, " is prime")
    i = i + 1

print ("Good bye!")
```

On executing, the above code produces following result –

```
2 is prime
3 is prime
5 is prime
7 is prime
11 is prime
13 is prime
17 is prime
19 is prime
23 is prime
Good bye!
```

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